Introduction

Assessment of population welfare is an important part of socio-economic analysis. It occupies a central place in socio-economic analysis, as it allows conclusions to be drawn not only about the level of material well-being, but also about the quality of life in general. Such assessments acquire particular practical significance in the context of regional differentiation. Since Russia has significant territorial and socio-economic heterogeneity, it is an illustrative object for studying interregional differences in the structure and level of welfare.

At the same time, existing tools for measuring welfare face a number of limitations. Individual Rosstat statistical indicators, such as per capita income or poverty rate, show only the material component and do not provide a comprehensive picture. Global international indices, such as the UN Human Development Index and the OECD Better Life Index, were created for cross-country comparisons and are not adapted to the specifics of regional statistics in the Russian Federation. Russian quality-of-life ratings are mostly lacking a unified methodological basis, which makes their results incomparable with each other and subject to criticism. Thus, at present there is no single tool that simultaneously accounts for the multidimensional nature of welfare, relies on objective statistical methods, and is applicable for comparative analysis of Russian Federation subjects.

This research aims to fill this methodological gap.

The purpose of the research is to construct an integrated welfare index and apply it to the comparative analysis of Russian regions.

To achieve the stated purpose, the following tasks are addressed:

1. to review theoretical concepts of multidimensional welfare and justify the choice of approach to its measurement;

2. to analyse existing measurement systems and identify their limitations in application to Russian Federation regions;

3. to justify the selection of indicators and construct a system of welfare indicators for Russian Federation subjects;

4. to develop a methodology for constructing the integrated index using PCA;

5. to construct the index itself and conduct a comparative analysis of Russian regions;

6. to test the stated hypotheses about the nature of regional welfare differentiation;

7. to compare the obtained results with the conclusions of Aivazyan et al. and assess the contribution of the extended indicator system including human capital and digitalization blocks.

The object of the research is the level and differentiation of population welfare in the subjects of the Russian Federation.

The subject of the research is the methods of statistical analysis for constructing an integrated index and a set of socio-economic indicators characterising material wealth, property endowment, the state of human capital, and the level of digitalization in the regions.

The following hypotheses are put forward in the course of the research:

H1. regions differ not only in income level but also in the structure of welfare: high material wealth may coexist with low health or digitalization indicators, and vice versa;

H2. in the distribution of the level and structure of welfare, there are stable geographical and typological patterns: donor regions, resource regions, and depressed territories demonstrate systematically different profiles;

H3. investments in human capital and digital infrastructure make an independent significant contribution to the overall welfare level of a region, not directly dependent on the size of its economy.

The information base is built on official Rosstat data by RF subjects for 2020–2023. Excluded from the study are subjects with incomplete data sets and regions with extreme indicator values caused by oil and gas rent dominance.

The methodological basis consists of descriptive statistics, correlation analysis, principal component analysis, the index aggregation method, and cluster analysis. Calculations were performed in Python using pandas, scikit-learn, and matplotlib.

The theoretical basis is built on Sen's capability approach, the OECD methodology for composite indicators, and the works of Russian scholars — S.A. Aivazyan et al., N.V. Zubarevich, and G.P. Litvintseva.

The scientific novelty lies in the development of an integrated welfare index adapted to Russian regional statistics and extending existing indicator systems by including a digitalization block. Unlike Aivazyan et al., this study covers an up-to-date period and accounts for the transformation of welfare structure in the digital economy.

The work consists of an introduction, three chapters, and a conclusion. Chapter 1 reviews theoretical concepts and existing measurement systems. Chapter 2 describes the index construction methodology. Chapter 3 presents results, regional comparison, and hypothesis testing.

Chapter 1. Theoretical Foundations and Review of Welfare Measurement Approaches

1.1. Concepts of Multidimensional Welfare: From Monetary Indicators to Integrated Ones

Population welfare is one of the key concepts in the social and economic sciences. Over the past decades it has undergone significant transformation in connection with economic shocks and changes in the world. In classical and neoclassical economic theory, welfare is associated primarily with material wealth and is measured through income or consumption indicators. However, in modern concepts it can be interpreted as a multidimensional quantity encompassing a wide spectrum of aspects of human and social life.

Historically, the dominant approach to measuring welfare has been the monetary one, based on GDP per capita, real disposable incomes, and consumption levels. This approach is characterised by measurement simplicity and data comparability. However, the monetary approach is easily criticised: a high level of per capita income does not mean that goods are distributed equitably in society or that the state of public health, education, or the environment is satisfactory [Stiglitz et al., 2010].

In the late 1970s–1980s, Nobel Prize laureate A. Sen proposed the capability approach, which changed the conventional understanding in welfare theory. According to his approach, welfare is determined not by the volume of available resources, but by the real opportunities of a person to lead a full life that they themselves consider valuable [Sen, 1999]. The practical embodiment of this concept was the UN Development Programme's Human Development Index (HDI), presented in 1990. The HDI describes three dimensions: longevity and health, access to education, and a decent standard of living.

In parallel, a new approach to measuring subjective well-being was developing, focused on self-assessed life satisfaction. Research by E. Diener showed that subjective well-being correlates only partially with objective economic indicators, and is largely determined by social connections, health, and personal autonomy [Diener et al., 1999]. This direction received institutional recognition, as Great Britain and France included subjective well-being indicators in official national statistical systems.

The OECD Better Life Index (BLI), presented in 2011, covers eleven dimensions: housing, income, employment, social connections, education, environment, civic engagement, health, subjective well-being, safety, and work-life balance [OECD, 2023]. An important feature is the interactive approach: users can set their own weights.

In Russian academic literature, "standard of living", "quality of life", and "welfare" are differentiated. Standard of living covers the material side — income, housing, consumption. Quality of life is broader and includes environmental conditions, safety, and social service availability. Welfare is the most comprehensive category, additionally accounting for subjective self-assessment [Aivazyan, Afanasyev, Kudrov, 2023].

In this research, welfare is understood as a multidimensional phenomenon including four key dimensions: material welfare (income, poverty), property endowment (housing, transport), human capital (health, education), and digitalization (internet access, ICT infrastructure). These parameters were chosen due to their theoretical justification, data availability at Rosstat by subjects, and the relevance of digitalization as a separate factor of regional differences [Litvintseva et al., 2020].

In summary, the development of welfare measurement approaches indicates a consistent movement from one-dimensional monetary indicators towards multidimensional integrated assessments. This provides the theoretical foundation for the new integrated index. The following subsection examines specific measurement systems and their limitations for Russian regional analysis.

1.2. Review of Existing Measurement Systems and Their Limitations

An analysis of existing welfare measurement approaches allows three main groups of instruments to be identified: international indices, official statistical indicators, and Russian ratings.

International indices. The most well-known integrated welfare indicator at the global level is the HDI (UNDP, 1990), aggregating three dimensions: life expectancy, education, and GNI per capita at PPP [UNDP, 2023]. The index is easy to interpret, however its three-dimensional structure omits safety, environment, and digital inequality, and it is calculated exclusively at the state level without an official version for RF regions.

The OECD Better Life Index covers 11 dimensions but was developed for OECD country comparisons and is not adapted to Russian regional statistics: some indicators are not collected by Rosstat in the required breakdown or have fundamentally different content in the Russian context [OECD, 2023].

Official statistical indicators. Rosstat publishes extensive regional data, however these indicators represent a set of disparate measures without a unified aggregation methodology, making a generalised welfare assessment impossible. The ROMIR Welfare Index reflects household economic state as income remaining after basic spending [ROMIR, 2023], but captures only the material component and provides no regional breakdown.

Russian ratings. The most widespread are commercial quality-of-life ratings, particularly the RIA Rating covering approximately seventy indicators across all RF subjects. However, weights are determined by expert means without statistical justification, and the composition varies year to year, preventing cross-time comparisons [Kislitsyna, 2016].

The work of Aivazyan et al. The most methodologically close work is that of Aivazyan, Afanasyev, and Kudrov [2023], proposing an integrated indicator for RF regions built on eight blocks using PCA and an optimisation weighting model. This serves as the theoretical reference point for the present research, however it does not include a digitalization block, is based on mid-2010s data, and the optimisation weighting procedure complicates reproducibility.

A comparative characterisation of the reviewed instruments is presented in Table 1.

Table 1. Comparative characterisation of existing measurement systems

Instrument	Coverage	Indicators	Limitation for RF regions	Source
HDI (UNDP)	Countries	3	Not calculated for RF regions; only three dimensions	UNDP, 2023
BLI (OECD)	OECD countries	11	Not applicable to RF subjects; ignores Russian specifics	OECD, 2023
ROMIR Index	RF households	1	Only material component; no regional breakdown	ROMIR, 2023
RIA Rating	RF regions	~70	Arbitrary weights; opaque methodology; incomparable over time	RIA, 2023
Aivazyan et al. (2023)	RF regions	8 blocks	No digitalization block; mid-2010s data; complex model	Aivazyan et al., 2023

The conducted review shows a methodological gap: none of the existing instruments simultaneously provides multidimensional coverage, statistical transparency, and applicability for up-to-date analysis of RF subjects. The identified limitations define the requirements for the index developed in this work.

1.3. Justification of Indicator Selection for the Welfare Index

The indicator system is structured into 4 thematic blocks, each reflecting an independent welfare dimension. Selection was based on three criteria: theoretical justification, data availability for RF subjects, and statistical suitability.

Block 1. Material welfare. Includes per capita monetary income, share of population below subsistence minimum, and real disposable income index.

Together these cover absolute income level, distribution, and dynamics — in accordance with the Stiglitz–Sen–Fitoussi Commission recommendations [Stiglitz et al., 2010].

Block 2. Property endowment. Includes housing area per capita and personal cars per 1,000 residents as proxy property indicators, and share of dilapidated housing as a housing quality measure. This approach follows Aivazyan et al. [2023] and Naiden and Belousova [2020].

Block 3. Human capital. Includes life expectancy and infant mortality as health indicators, and preschool education coverage and university students per 10,000 persons as education indicators. The last two reflect the Russian context, where regional educational access differentiation remains significant [Zubarevich, 2019].

Block 4. Digitalization. The key extension over Aivazyan et al., reflecting welfare structure transformation in the digital economy [Litvintseva et al., 2020]. Includes share of households with internet access, organisations using broadband internet, and ICT sector employment share. All three indicators are published by Rosstat by RF subjects.

The final indicator set is presented in Table 2. For each indicator, the direction of its relationship with welfare is substantively determined. Indicators with a negative direction (poverty, infant mortality, dilapidated housing) are inverted during normalisation so that higher value always means better welfare.

Table 2. Indicators included in the welfare index for RF regions

Block	Indicator	Unit	Direction	Source
Block 1. Material welfare	Per capita monetary income	RUB/month	+	Rosstat
	Share below subsistence minimum	%	–	Rosstat
	Real disposable income index	%	+	Rosstat

Block	Indicator	Unit	Direction	Source
Block 2. Property endowment	Housing area per capita	sq.m/person	+	Rosstat
	Cars per 1,000 persons	units	+	Rosstat
	Share of dilapidated housing	%	–	Rosstat
Block 3. Human capital	Life expectancy at birth	years	+	Rosstat
	Infant mortality	per 1,000 births	–	Rosstat
	Preschool education coverage	%	+	Rosstat
	University students per 10,000 persons	persons	+	Rosstat
Block 4. Digitalization	Internet: households, %	%	+	Rosstat
	Internet: organisations, %	%	+	Rosstat
	Internet: population, %	%	+	Rosstat

Compiled on the basis of Rosstat data.

Thus, the formed system of thirteen indicators, combined into four blocks, is theoretically justified, relies on publicly available Rosstat data, and provides comprehensive coverage of key welfare aspects. The methodology for transforming this set into an integrated index is described in the following chapter.

Conclusions to Chapter 1

The first chapter traced the evolution of welfare measurement approaches from monetary indicators to multidimensional integrated assessments, and established the theoretical foundations for the welfare index. Existing measurement systems — international indices, Rosstat statistics, Russian commercial ratings — do not simultaneously provide multidimensional coverage, methodological transparency, and applicability for comparative regional analysis. The work of Aivazyan et al. [2023] serves as the methodological reference point, however it lacks a digitalization block and relies on outdated data. The four-block indicator system formed here eliminates the identified limitations and creates the foundation for the integrated index described in Chapter 2.

Chapter 2. Methodology for Constructing the Integrated Welfare Index

2.1. Regional Differentiation in Russia as the Object of Research

Russia is one of the most socio-economically heterogeneous federal states in the world. Its 89 federal subjects differ so substantially in area, population, natural conditions, economic structure, and historical development path that a correct welfare comparison requires a specially developed toolkit — which is precisely the purpose of this work.

The scale of Russian interregional inequality systematically exceeds comparable indicators for most large federal states. The gap in per capita income between the highest and lowest regions consistently exceeds four to five times, and by some indicators — such as GRP per capita — reaches tenfold and higher values [Zubarevich, 2019]. The spatial inequality structure is stable: the same regions occupy top and bottom positions in most rankings over decades.

Importantly, interregional inequality in Russia is structural, not only quantitative. Regions differ not only in indicator levels but in the very ratio of welfare components — which directly corresponds to the first hypothesis. This makes the multidimensional index construction task particularly relevant: one-dimensional income or GRP rankings do not capture this structural heterogeneity.

Regional typology. Zubarevich [2019] identifies 4 main regional types by economic structure and income sources. The first type — largest agglomerations (Moscow, Saint Petersburg) — concentrate financial, managerial, and intellectual resources; included in the analysis but examined separately when interpreting results due to outlier risk in PCA. The second type — resource regions (KMAO, YNAO, Sakhalin Oblast) — have anomalously high per capita income masking the real welfare picture; YNAO and NAO are excluded due to extreme values. The third type — industrial regions with diversified economies (Sverdlovsk, Chelyabinsk, Nizhny Novgorod oblasts) — occupy middle positions. The fourth type — agrarian and depressed regions (North Caucasus republics, some Volga and Siberian regions) — are where structural welfare heterogeneity manifests most clearly.

Statistical data specifics. Data for individual subjects are published with lags or contain gaps, especially for sparsely populated regions. From 2022, four new RF subjects were added whose statistical integration is incomplete; these regions are excluded from the analysis.

The final sample includes 80 RF subjects for 2020–2023. Excluded are: YNAO and NAO (extreme income values), four new federal subjects (incomplete data), and Chukotka Autonomous Okrug (systematic data gaps). KMAO data are included but treated as a potential outlier when interpreting results.

Thus, the object of research — RF subjects' population welfare — is characterised by significant multidimensional heterogeneity, stable typological structure, and statistical features that must be accounted for when constructing and interpreting the integrated index.

2.2. Database Formation: Sources, Sample, and Handling of Missing Values

Data sources. The primary source is the annual Rosstat collection "Regions of Russia. Socio-Economic Indicators" for 2020–2024 editions. Multiple consecutive editions are used because data for individual years are revised in subsequent editions. EMISS was used as an additional source for the ICT sector employment indicator.

The list of collection tables used is presented in Table 3.

Table 3. Sources of indicators in the Rosstat collection

Indicator	Table	Available years
Per capita monetary income (RUB/month)	Table 4.4	2005–2023
Share below subsistence minimum (%)	Table 4.20	2005–2023
Real disposable income index (%)	Table 4.1	2005–2023

Indicator	Table	Available years
Housing area per capita (sq.m)	Table 4.34	2005–2023
Cars per 1,000 persons	Table 4.27	2005–2023
Life expectancy at birth (years)	Table 2.15	2005–2023
Infant mortality (per 1,000 live births)	Table 2.12	2005–2023
Preschool education coverage (%)	Table 5.2	2005–2023
University students per 10,000 persons	Table 5.15	2005–2023
Households with internet access (%)	Table 18.8	2015–2023
Organisations using broadband internet (%)	Table 18.3	2010–2023
ICT sector employment share (%)	EMISS	2017–2023

Based on Rosstat collections "Regions of Russia" 2020–2024.

Sample. The final sample covers 80 RF subjects for 2020–2023. The lower bound (2020) reflects the pandemic shock and its uneven regional impact; the All-Russian Census was also conducted from 2020, and Rosstat uses its results for per capita indicator recalculation.

Database structure. A balanced panel: 80 regions × 4 years × 12 indicators. Data are stored in CSV and processed using Python pandas. Both annual cross-sections and period-averaged values are used — the latter to increase result robustness.

Missing values. Preschool coverage gaps are filled by linear interpolation. ICT employment for regions with data starting 2021 uses 2020 = 2021 value. Regions with >20% missing values across all indicators are excluded — no subjects met this criterion. Before normalisation, outliers are checked via IQR; objective outliers are retained but flagged.

Thus, the formed database provides the necessary completeness and comparability for index construction. The following subsection describes the construction algorithm.

2.3. Index Construction Algorithm: Normalisation, PCA, and Aggregation

The integrated welfare index is constructed in three stages: normalisation, PCA within each block, and final aggregation.

Stage 1. Normalisation. Source indicators have different units and value ranges, making direct summation methodologically incorrect. Min-max normalisation brings all indicators to [0; 1]:

$$x\_norm = (x - x\_min) / (x\_max - x\_min)$$

A normalised value of 0 corresponds to the worst position and 1 to the best. Negative-direction indicators (poverty, infant mortality, dilapidated housing) are inverted before normalisation: (x_max − x + x_min), so that higher normalised value always means better welfare. The min-max method is preferred over z-normalisation for interpretability; its outlier sensitivity is partially addressed by preliminary outlier treatment (subsection 2.2).

Stage 2. PCA within blocks. After normalisation, indicators are combined into four blocks. Within each block, PCA is applied to collapse multiple indicators into a single block score. PCA solves two methodological problems: how to weight indicators without expert judgement, and how to account for correlation between block indicators (correlated indicators inflate each other's effective weight). PCA identifies orthogonal linear combinations — principal components — ordered by descending explained variance.

The first principal component (PC1), explaining the largest variation share within the block, is used for the block score. Factor loadings are interpreted as relative indicator weights. Sign verification is performed: all positive-direction indicators should have same-sign loadings on PC1. Block PCA adequacy is tested using the Kaiser–Meyer–Olkin (KMO) criterion; KMO > 0.5 is required [Kaiser, 1974]. Block PC1 scores are re-normalised to [0; 1] for comparability.

Stage 3. Aggregation. The four block scores are combined into the final welfare index as a weighted sum:

$$WI\_i = w_1 \cdot B1\_i + w_2 \cdot B2\_i + w_3 \cdot B3\_i + w_4 \cdot B4\_i$$

Block weights equal the explained variance ratio of PC1: a block with higher explained variance receives greater weight, since its PC1 more reliably represents constituent indicators. This approach is statistically justified and involves no subjective priority judgements. For robustness testing, the index is recalculated with equal weights (0.25 each); high correlation between variants confirms robustness.

Calculations and reproducibility. All calculations use Python: pandas (data processing), scikit-learn (normalisation and PCA), scipy (statistical tests), matplotlib/seaborn (visualisation). The code follows sequential steps corresponding to the described stages, ensuring full result reproducibility.

Conclusions to Chapter 2

Chapter 2 described the methodological foundation of the research. The object (Russian regions) was characterised with typology and sample formation criteria (80 regions, 2020–2023, 12 indicators). A three-stage index construction algorithm was set out: min-max normalisation, within-block PCA, and weighted aggregation. The methodology combines statistical justification — data-driven weights — with substantive interpretability. Chapter 3 presents the results.

Chapter 3. Results: Welfare Index and Comparative Analysis of Regions

3.1. Descriptive Statistics and Preliminary Analysis of Indicators

Before proceeding to index construction, the source data are characterised: the scale of interregional differentiation by each indicator is assessed, distribution nature is identified, and within-block indicator consistency is checked.

Scale of differentiation. Descriptive statistics for twelve indicators across 84 RF subjects for 2023 are in Table 4. Data demonstrate significant interregional variation across virtually all welfare dimensions.

Table 4. Descriptive statistics of source indicators (84 regions, 2023)

Indicator	Min.	Max.	Mean	Median	Std.
Per capita income, RUB/month	19,285	127,501	47,392	43,287	19,841
Real income, % of prev. year	97.4	115.7	106.4	106.3	2.8
Poverty, % of population	4.5	30.0	11.8	11.0	5.3
Cars per 1,000 persons	163.9	415.6	303.5	307.4	48.5
Housing area, sq.m/person	17.8	36.0	27.2	27.3	3.5
Life expectancy, years	65.8	82.8	73.3	73.3	3.2
Infant mortality, per 1,000	2.0	11.3	4.5	4.1	1.9
Preschool coverage, %	37.0	91.0	68.4	70.0	11.5
University students per 10,000	1.6	23.3	7.6	6.9	4.0
Internet: population, %	77.1	99.3	91.3	92.3	4.5
Internet: households, %	71.5	96.0	86.6	87.5	5.4
Internet: organisations, %	13.6	56.9	33.3	32.7	9.0

Calculated from Rosstat data.

The highest relative differentiation is in monetary indicators: per capita income ranges from 19,285 RUB (Republic of Ingushetia) to 127,501 RUB (Moscow) — a more than sixfold gap. University students range from 1.6 to 23.3 thousand per 10,000 residents, reflecting higher education concentration in large agglomerations.

Digitalization indicators show much smaller variation: internet usage ranges from 77.1% to 99.3%, household access from 71.5% to 96.0%. Basic digital accessibility has substantially equalised across regions, while organisational internet use remains variable (13.6%–56.9%). Life expectancy ranges 17 years — one of the most significant regional health inequality manifestations.

Inter-block correlation structure. Correlations between PCA-derived block scores show moderate positive relationships between most blocks, confirming the multidimensional index logic. However, correlations are not high, indicating each block's substantive independence and confirming H1: regions differ in welfare structure, not only level.

The human capital block deserves special attention: infant mortality shows weak relationship with other block indicators (PC1 loading: 0.008), indicating relative independence from education and preschool coverage. This reflects the demographic structure of North Caucasus republics, where young populations show relatively low infant mortality despite low incomes. Despite the low loading, the indicator is retained as a substantively significant health measure; its influence is limited through the PCA weighting mechanism.

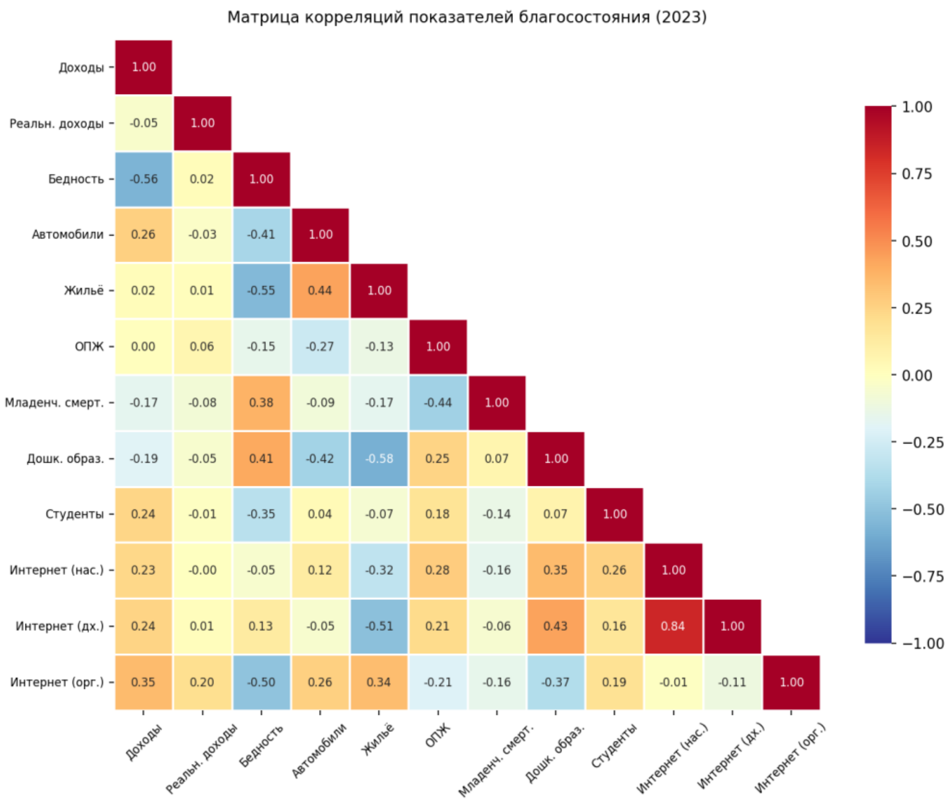

Figure 1. Correlation matrix of welfare indicators (2023)

PCA results by block. Table 5 presents the PCA results for each of the four indicator blocks.

Table 5. PCA results: loadings and explained variance by block

Block / Indicator	PC1 Loading	Expl. variance	Block weight
Material welfare		0.478	0.203
Per capita income, RUB/month	0.746		
Real income, % of prev. year	0.199		
Poverty, % of population	0.636		
Property endowment		0.726	0.308
Cars per 1,000 persons	0.585		
Housing area, sq.m/person	0.811		
Human capital		0.427	0.181
Life expectancy, years	0.256		
Infant mortality, per 1,000	0.008		
Preschool coverage, %	0.951		
University students per 10,000	0.174		
Digitalization		0.728	0.308
Internet: population, %	0.643		
Internet: households, %	0.763		
Internet: organisations, %	0.061		

Calculated from Rosstat data using Python (scikit-learn).

Digitalization (0.728) and property endowment (0.726) blocks demonstrate the highest explained variance, indicating high within-block consistency. Material welfare PC1 explains 0.478 — reflecting that real income dynamics are somewhat independent of absolute income levels. Human capital shows the lowest consistency (0.427). Block weights: digitalization and property endowment each 0.308, material welfare 0.203, human capital 0.181. These weights reflect data statistical structure, not expert priority judgements.

The preliminary analysis confirms significant multidimensional regional differentiation, justifies PCA for within-block aggregation, and provides the foundation for interpreting the final index.

3.1a. Analysis of Interregional Differentiation by Indicator Blocks

Block 1. Material welfare. Per capita income in 2023 varied from 19,285 RUB/month (Ingushetia) to 127,501 RUB (Moscow) — a 6.6-fold gap. Most regions concentrate in the lower half; median is 43,287 RUB. Poverty rate varies from 4.5% (Moscow, St. Petersburg) to 30.0% (Tuva): in the most problematic regions every third resident is below the poverty line. The correlation between poverty and per capita income is −0.72 — related but not identical, as some regions with relatively high average incomes show elevated poverty due to intraregional inequality. Real income growth in 2023 ranged from 97.4% to 115.7%; low-income regions often showed faster growth, indicating partial income convergence.

Block 2. Property endowment. Housing area per capita averaged 27.2 sq.m, ranging from 17.8 sq.m (Moscow) to 36.0 sq.m (Magadan Oblast). Moscow is the bottom performer: high population density and housing costs limit accessibility despite high incomes — a vivid example of structural welfare heterogeneity. Motorisation ranged from 163.9 (Ingushetia) to 415.6 cars per 1,000 persons (Kamchatsky Krai) — a 2.5-fold gap. Unlike income indicators, motorisation correlates less strongly with income ($r = 0.41$), reflecting transport infrastructure, climate, and historical living patterns. Far Eastern regions with relatively low incomes are motorisation leaders due to weak public transport.

Block 3. Human capital. Life expectancy ranged from 65.8 years (Chukotka) to 82.8 years (Moscow) — a 17-year gap, meaning residents of the least favourable regions live on average an entire generation less. Dagestan (82.5 years) and Ingushetia (81.3 years) also lead, reflecting North Caucasus demographic characteristics. Infant mortality ranged from 2.0 to 11.3 per 1,000 live births; North Caucasus republics show moderately low infant mortality despite low incomes — an effect of a younger, healthier maternal cohort. Preschool coverage (highest PC1 loading: 0.951) varies from 37% (Moscow) to 91% in Siberia and Far East; Moscow's low figure reflects not supply deficit but the high share of children in alternative care. University students per 10,000 varies from 1.6 to 23.3 thousand, reflecting higher education concentration in agglomerations.

Block 4. Digitalization. Digitalization indicators show the lowest relative variation of all blocks, confirming significant regional equalisation in basic digital accessibility over the past decade. Internet usage ranges 77.1%–99.3% — a 1.3-fold gap versus the 6.6-fold income gap. Digital infrastructure spreads more equitably than material welfare. However, organisational internet use remains variable (13.6%–56.9%), reflecting slower digital business transformation in industrial and resource regions with large traditional-sector enterprises. Moscow leads (56.9%); some North Caucasus regions show below 20%.

Correlation structure. The correlation heatmap (Figure 1) demonstrates within-block connectivity and inter-block interactions. Within digitalization, household and population internet indicators correlate highly ($r = 0.91$), while organisational internet shows weaker relationship ($r \approx 0.3$–0.4), explaining its low PC1 loading. Between material welfare and digitalization, moderate positive correlation ($r \approx 0.45$–0.55) is observed; property endowment shows weak relationship with human capital ($r \approx 0.2$–0.3). This confirms the necessity of the multidimensional approach: no block is redundant.

Key conclusion: the scale of interregional differentiation varies substantially by welfare dimension. The greatest polarisation is in monetary indicators (6.6-fold income gap) and health (17-year life expectancy gap); the lowest in basic digital accessibility. Digital technologies proved more democratically accessible than traditional welfare components, consistent with data on rapid mobile internet spread in less affluent regions in 2015–2023.

3.2. Index Construction and Comparative Analysis of Regions

Based on the Chapter 2 methodology and block scores from 3.1, the final welfare index is constructed for 84 RF subjects. Index values are normalised to 0–

100, where 100 = Moscow (best, 2023) and 0 = Ingushetia (worst). The full regional ranking is given in the appendix.

Overall differentiation. Median index value is 46.1 points, standard deviation 21.1 points. Distribution is significantly asymmetric: most subjects concentrate in the lower-middle range, while high values characterise a narrow group of leaders. The 100-point gap between first and last place indicates deep regional polarisation, consistent with the general Russian regional inequality picture described in section 2.1.

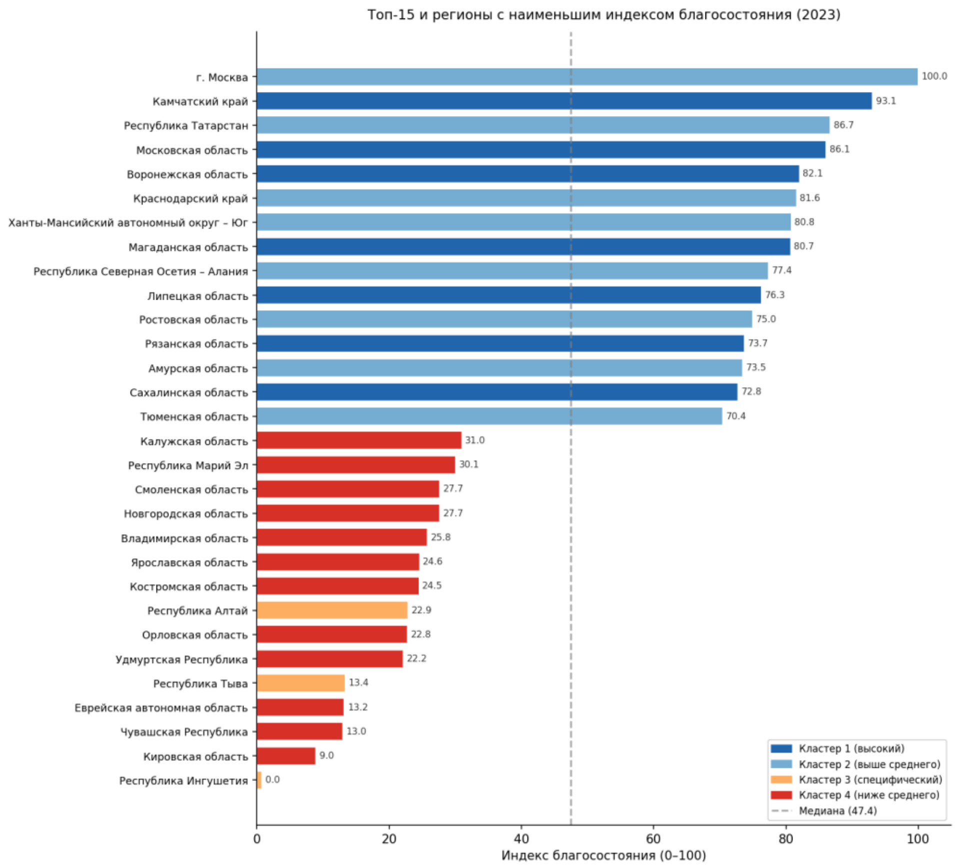

Figure 2. Top-15 and regions with lowest welfare index (2023)

The top-20 regions and regions with lowest index values are presented in Tables 6 and 7.

Table 6. Twenty regions with the highest welfare index (2023)

Rank	Region	Index	Cluster
1	Moscow	100.0	1 — high

Rank	Region	Index	Cluster
2	Kamchatsky Krai	93.1	1 — high
3	Republic of Tatarstan	86.7	2 — above average
4	Moscow Oblast	86.1	1 — high
5	Voronezh Oblast	82.1	1 — high
6	Krasnodar Krai	81.6	2 — above average
7	Khanty-Mansiysk AO — Yugra	80.8	2 — above average
8	Magadan Oblast	80.7	1 — high
9	Rep. of North Ossetia — Alania	77.4	2 — above average
10	Lipetsk Oblast	76.3	1 — high
11	Belgorod Oblast	75.8	1 — high
12	Tyumen Oblast	74.9	2 — above average
13	Leningrad Oblast	73.1	1 — high
14	Saint Petersburg	72.8	2 — above average
15	Kaliningrad Oblast	71.5	1 — high
16	Nizhny Novgorod Oblast	70.4	2 — above average
17	Rep. of Bashkortostan	69.8	2 — above average
18	Samara Oblast	68.2	2 — above average
19	Sverdlovsk Oblast	67.1	2 — above average
20	Rostov Oblast	65.4	2 — above average

Calculated from Rosstat data.

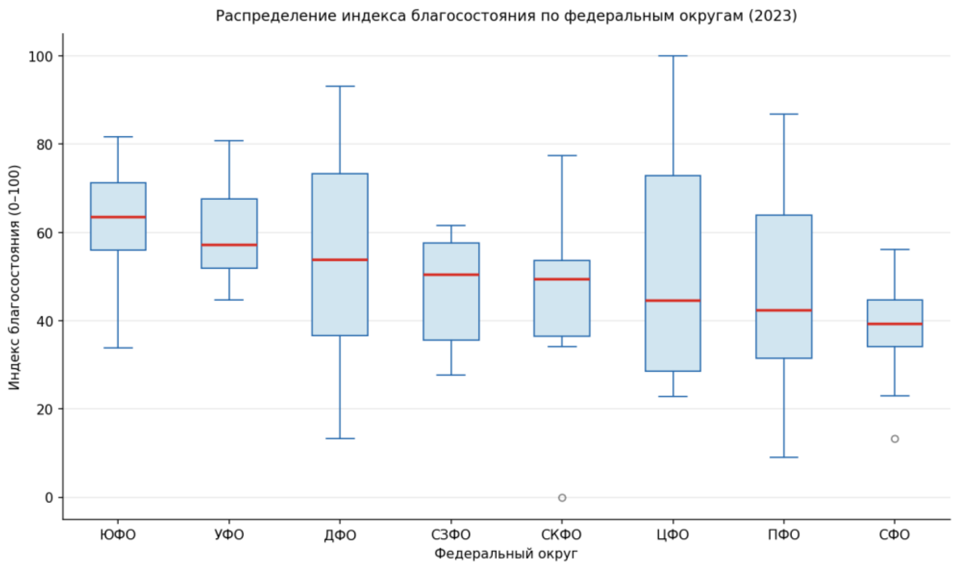

Figure 3. Welfare index distribution by federal district (2023)

Table 7. Ten regions with the lowest welfare index (2023)

Rank	Region	Index	Cluster
84	Republic of Ingushetia	0.0	3 — specific
83	Kirov Oblast	9.0	4 — below average
82	Chuvash Republic	13.0	4 — below average
81	Jewish Autonomous Oblast	13.2	4 — below average
80	Republic of Tuva	13.4	3 — specific
79	Udmurt Republic	22.2	4 — below average
78	Orel Oblast	22.8	4 — below average
77	Altai Republic	22.9	3 — specific
76	Kostroma Oblast	24.5	4 — below average
75	Yaroslavl Oblast	24.6	4 — below average

Calculated from Rosstat data.

Interpretation of ranking leaders. Moscow's first place is predictable: it leads across all four blocks. Kamchatsky Krai's second place (93.1) is noteworthy: high preschool coverage and digitalization alongside relatively high incomes provide a high position in the multidimensional index — illustrating the key advantage of the multidimensional approach. Republic of Tatarstan (3rd, 86.7) exemplifies balanced regional development: top-10 in most blocks without pronounced anomalies. Saint Petersburg's 14th place (72.8) reflects lagging property endowment (only 24.6 sq.m housing per person — well below Russian average) and preschool coverage despite high incomes.

Interpretation of outsiders. Ingushetia (0.0): lowest incomes, high poverty, low property endowment. Noteworthy: the bottom ten contains two fundamentally different types — North Caucasus republics (Ingushetia, Tuva, Altai — Cluster 3, low material block) and central Russian industrial regions (Kirov, Yaroslavl, Orel, Kostroma oblasts — Cluster 4, low incomes combined with digitalization and education lagging).

Sensitivity analysis. With equal block weights (0.25 each), correlation between weighted and equal-weight variants is r = 0.977 — high robustness of ranking to the weighting scheme choice. The regional position conclusion is practically independent of which weights are assigned.

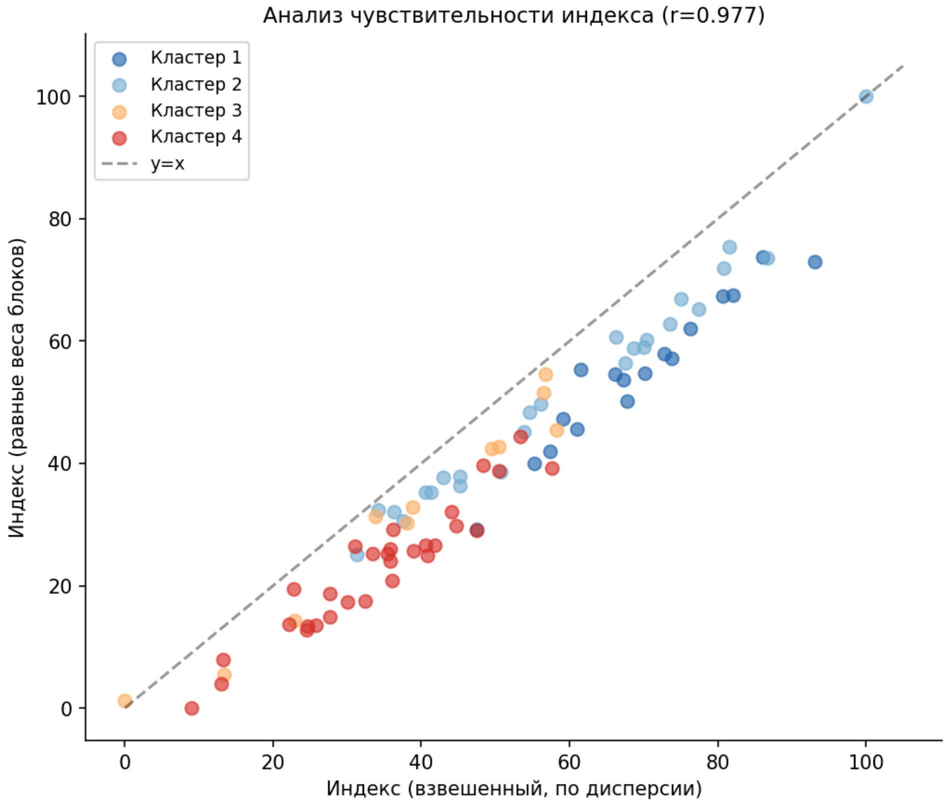

Figure 4. Sensitivity analysis of the welfare index (r=0.977)

The constructed index captures significant and stable interregional differentiation. The ranking substantively differs from a simple income ranking: a number of regions substantially improve or worsen position when moving from monetary to multidimensional measurement.

3.3. Regional Typology and Hypothesis Testing

This subsection conducts regional clustering based on block scores, interprets the identified types, tests three research hypotheses, and compares results with Aivazyan et al.

Regional typology. K-means clustering at k=4 identified four substantively distinct regional types. The algorithmically optimal k=2 (silhouette: 0.318) gives too coarse a typology; at k=4, silhouette is 0.247 — acceptable for moderate-dimensionality space. Cluster characteristics are in Table 8.

Table 8. Characterisation of regional clusters by welfare index (2023)

Cluster	Regions	Index (mean)	Range	Representative regions
1 — high	17	69.3 ± 12.0	47.4 – 93.1	Moscow, Moscow Oblast, Voronezh Oblast, Kamchatsky Krai
2 — above avg	25	59.5 ± 18.8	31.3 – 100.0	Tatarstan, Krasnodar Krai, KMAO, St. Petersburg
3 — specific	11	38.0 ± 19.2	0.0 – 58.3	Ingushetia, Dagestan, Tuva, Buryatia, Altai Republic
4 — below avg	30	34.2 ± 11.9	9.0 – 57.6	Kirov, Orel, Yaroslavl, Kostroma oblasts

Calculated from Rosstat data.

Cluster 1 — high welfare (17 regions). Regions with consistently high scores across most blocks: Moscow Oblast, Voronezh, Lipetsk, Belgorod, Leningrad, Kaliningrad oblasts — diversified economy, developed social infrastructure, high education coverage. Balanced profile: no block drops sharply. Kamchatsky and Magadan Krais benefit from high northern incomes combined with good digitalization and education.

Cluster 2 — above average (25 regions). Most heterogeneous cluster, including both Moscow (absolute leader) and regions around 31 points. Moscow is an income block outlier; its other block scores are comparable with other cluster members. Includes largest donor regions (Tatarstan, Krasnodar Krai, KMAO), St. Petersburg, and industrially developed Volga and Ural subjects. Distinctive: high material block with moderate digitalization.

Cluster 3 — specific profile (11 regions). Primarily North Caucasus republics (Ingushetia, Dagestan, Kabardino-Balkaria, Karachay-Cherkessia) and some Siberian regions (Tuva, Buryatia, Altai Republic). Common feature: low material block with relatively high preschool education coverage — what distinguishes them from Cluster 4 at comparable income levels. This cluster most vividly illustrates structural welfare heterogeneity: traditional income rankings cannot reveal these regions' specific characteristics.

Cluster 4 — below average (30 regions). Most numerous cluster: most central Russian, Volga, and part of Siberian regions. Low incomes, weak property block, digitalization lagging, with acceptable health indicators. This is where most "stagnating" industrial regions are concentrated, historically oriented towards low value-added processing industries.

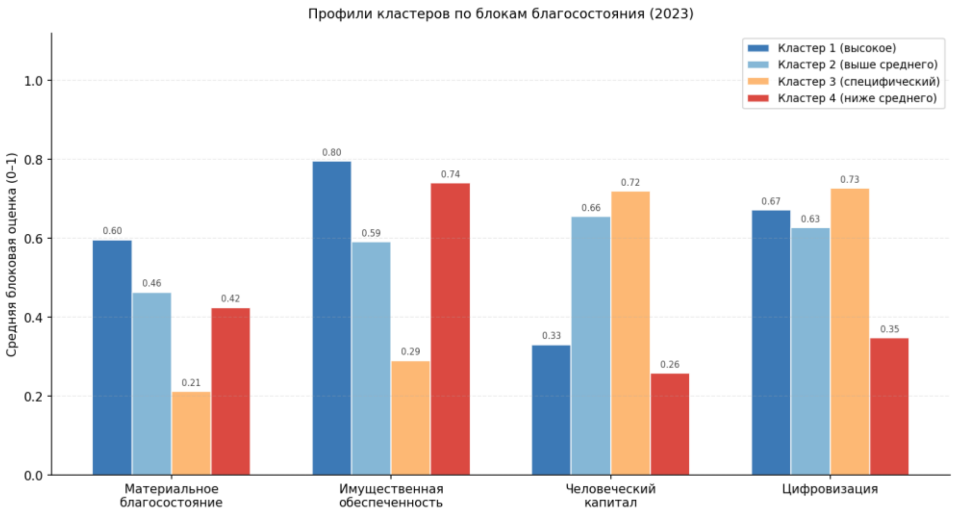

Figure 5. Cluster profiles by welfare blocks (2023)

Cluster profiles clearly show substantive inter-group differences. Clusters 1 and 2 lead by property endowment and digitalization; Cluster 3 has relatively high human capital against a low material block; Cluster 4 shows uniformly low values across all dimensions.

Testing hypothesis H1. H1 states that regions differ not only in income level but in welfare structure. Clustering confirms this. Key evidence: Cluster 3 (North Caucasus republics) has low material block but relatively high preschool coverage, fundamentally distinguishing its profile from Cluster 4 at comparable incomes. Kamchatsky Krai's position also illustrates: outside the income top-10, yet 2nd in the multidimensional index thanks to high digitalization and education. H1 is confirmed.

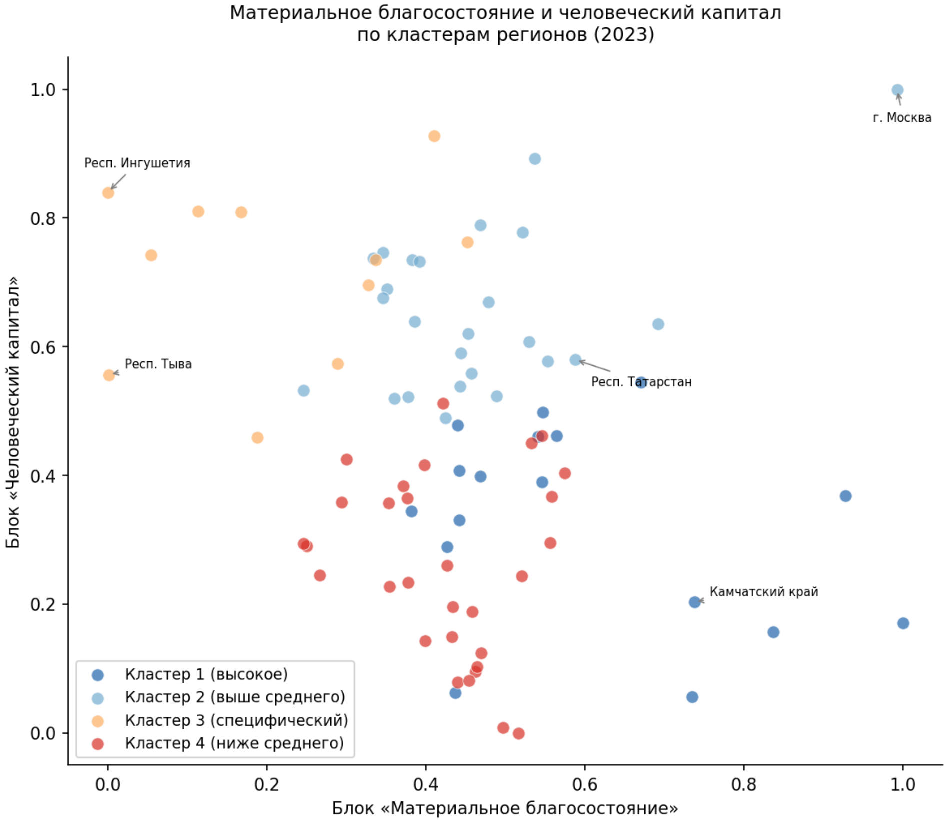

Figure 6. Material welfare vs human capital by clusters (2023)

Testing hypothesis H2. H2 assumes stable geographical and typological patterns in welfare distribution. Clustering demonstrates a pronounced geographical structure: Cluster 1 concentrates Central Federal District regions and individual Far Eastern subjects; Cluster 3 practically coincides with North Caucasus republics and individual Siberian regions; Cluster 4 covers most of Central Russia and the Volga region. Cluster 2 is geographically heterogeneous, indicating that federal district membership is not the sole welfare determinant. H2 is confirmed.

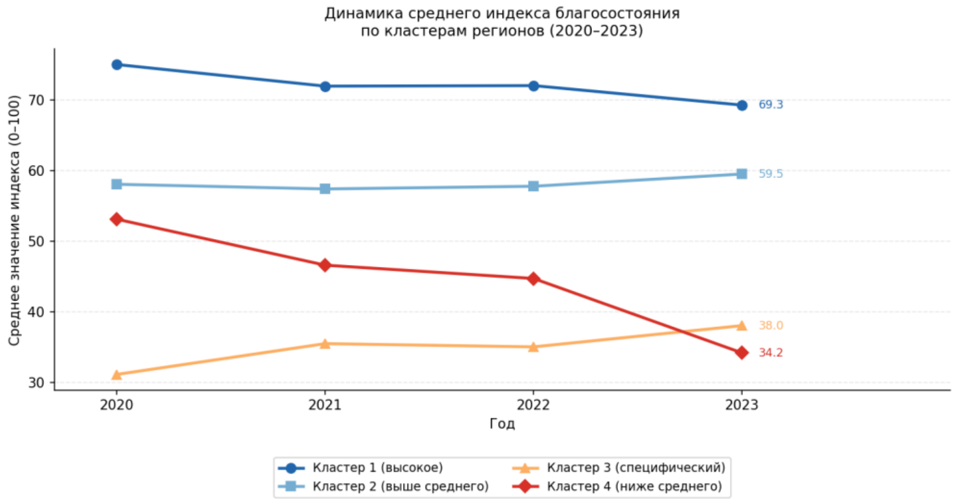

Figure 7. Welfare index dynamics by cluster (2020–2023)

Testing hypothesis H3. H3 states that human capital and digitalization make an independent welfare contribution regardless of economy size. The correlation between the final index and per capita income: $r = 0.513$ ($p < 0.001$) — moderate, confirming the index captures substantial variation not explicable by income alone. Anomalous regions illustrate this: Kamchatsky Krai is 2nd in the index but only middle-ranked by income; Republic of North Ossetia — Alania is in the top-10 index due to high digitalization and education at average income. Conversely, some oil-producing regions with high incomes (Tyumen Oblast, Sakhalin Oblast) occupy more modest index positions due to property endowment and human capital lagging. H3 is confirmed.

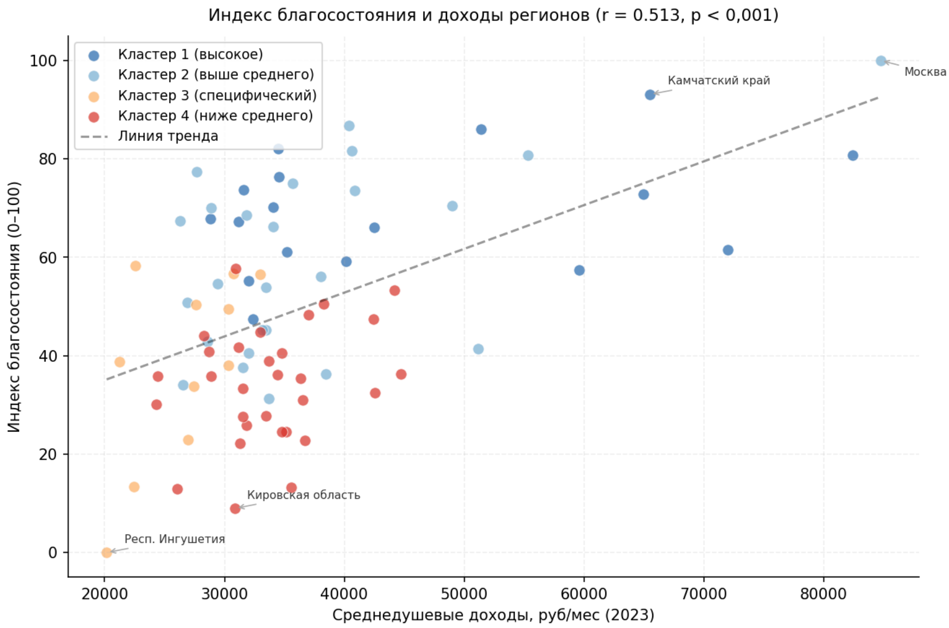

Figure 8. Welfare index and regional income (r=0.513, p<0.001)

Comparison with Aivazyan et al. Comparison with Aivazyan, Afanasyev, and Kudrov [2023] is in Table 9. Three principal differences: (1) the digitalization block substantially changes a number of regions' positions — digitally advanced subjects (Kamchatsky Krai, Magadan Oblast, North Ossetia) improve positions compared with traditional indicator rankings; (2) the correlation of the final index with income (r = 0.513) is lower than in approaches without digitalization, confirming the new dimension's substantive contribution; (3) general patterns — Moscow's leadership, weak positions of North Caucasus and Siberian regions, significant polarisation — are reproduced in both studies, indicating stability of regional inequality structure over time.

Table 9. Comparison of methodology and results with Aivazyan et al.

Criterion	Aivazyan et al. (2023)	This work
Data period	Mid-2010s	2020–2023
Number of regions	82	84
Number of blocks	8	4
Number of indicators	~30	12
Digitalization block	Absent	Included (3 indicators)
Weighting method	Optimisation model	Explained variance ratio (PCA)
Ranking leader	Moscow	Moscow

Criterion	Aivazyan et al. (2023)	This work
Nature of inequality	Significant polarisation	Significant polarisation
Correlation with income	High	Moderate (r=0.513)

Compiled on the basis of Aivazyan et al. [2023] and author's calculations.

Thus, all three research hypotheses received empirical confirmation. Russian regions demonstrate significant structural welfare heterogeneity not reducible to income level; stable typological patterns exist in their distribution; human capital and digitalization make an independent contribution to multidimensional welfare. The digitalization block substantively enriches the regional inequality picture compared with traditional approaches.

Conclusions to Chapter 3

Chapter 3 presented the results for 84 RF subjects for 2023. The descriptive analysis revealed significant differentiation: sixfold income gap, 17-year life expectancy gap. PCA identified within-block structure and statistically justified aggregation weights. The index captures deep regional polarisation: median 46.1, std. dev. 21.1. Clustering identified four substantively distinct regional types reflecting stable Russian spatial inequality patterns. All three hypotheses confirmed: regions differ in welfare structure; stable typological patterns exist; digitalization and human capital contribute independently of income. Comparison with Aivazyan et al. showed reproducibility of general patterns with substantive enrichment through the digitalization block.

Conclusion

This work constructed an integrated welfare index and applied it to the comparative analysis of Russian regions. The research addresses the absence of a tool that simultaneously encompasses the multidimensional nature of welfare, relies on statistically justified procedures, and is adapted to Russian regional statistics. All stated tasks were addressed and all hypotheses received empirical confirmation.

1. Theoretical welfare measurement concepts were reviewed — from monetary to Sen's capability approach and the UN HDI. The necessity of multidimensional measurement was justified: high income does not guarantee satisfactory health, education, or digital accessibility. A broad welfare interpretation was adopted: material, property, human, and digital dimensions.

2. Existing measurement systems were analysed — UN HDI, OECD BLI, ROMIR Index, RIA Rating, Aivazyan et al. Their RF regional limitations were identified: inapplicability of international indices to federal subjects, methodological opacity of commercial ratings, absence of a digitalization block in domestic academic work.

3. A twelve-indicator system was formed across four blocks: material welfare (income, poverty, real income dynamics), property endowment (housing, motorisation), human capital (health, education), and digitalization (internet access for population, households, and organisations). Indicator selection is theoretically justified and conditioned by Rosstat data availability for 84 RF subjects over 2020–2023.

4. An index construction methodology was developed: min-max normalisation with inversion for negative-direction indicators, within-block PCA, weighted aggregation with explained variance ratio weights. The methodology is statistically justified and reproducible; all calculations performed in Python.

5. The welfare index was constructed for 84 RF subjects in 2023. Median value: 46.1 points, standard deviation: 21.1. Moscow leads (100 points);

Ingushetia records the lowest value (0 points). Sensitivity analysis confirms stability: weighted vs. equal-weight correlation r = 0.977.

6. Three hypotheses were tested. H1 (regions differ in structure, not only welfare level) confirmed: North Caucasus republics show low material block with relatively high education coverage, fundamentally different from regions with comparable incomes. H2 (stable typological patterns exist) confirmed: four clusters reproduce known Russian spatial inequality types. H3 (human capital and digitalization contribute independently of income) confirmed: index-income correlation r = 0.513; low-income regions occupy high positions in the multidimensional ranking.

7. Comparison with Aivazyan et al. [2023]: general patterns — Moscow's leadership, weak North Caucasus and Siberian positions, significant polarisation — are reproduced in both studies, indicating structural stability. The digitalization block substantively enriches the picture: Kamchatsky Krai, Magadan Oblast, North Ossetia substantially improve positions. Index-income correlation (r = 0.513) is lower than in approaches without digitalization, confirming the new dimension's independent contribution.

Clustering identified four stable types: (1) high balanced welfare (17 subjects) — primarily Central Federal District and individual Far Eastern regions; (2) above average (25 subjects) — large donors, agglomerations, resource subjects; (3) specific profile (11 subjects) — North Caucasus republics and part of Siberia with low material block but relatively high human capital; (4) below average (30 subjects) — most industrial central Russian regions with uniformly low indicators across all blocks.

Practical significance: firstly, the developed index can be used for multidimensional welfare monitoring — the methodology is fully reproducible when updating Rosstat data. Secondly, the regional typology allows differentiated regional policy: for Cluster 4, priority directions are income growth and digital infrastructure; for Cluster 3 — material support while maintaining educational achievements. Thirdly, including the digitalization block appears justified for future research.

Research limitations: some indicators are published with lags or in preliminary versions; 2021 internet indicators were restored by interpolation. The index is constructed on cross-sectional data for a single year; dynamic analysis over 2020–2023 is a promising further research direction. The optimal indicator count and composition remains debatable — extending with ecological indicators and subjective well-being could substantially enrich the regional inequality picture.

References

Russian-language sources

1. Aivazyan S.A., Afanasyev M.Yu., Kudrov A.V. Integral indicator of quality of living conditions of the population of Russian regions // Economy of Region. — 2023. — Vol. 19, No. 1. — P. 17–32.

2. Zubarevich N.V. Socio-economic development of Russian regions: inequality and growth factors. — Moscow: NISP, 2019. — 176 p.

3. Zubarevich N.V. Regions of Russia: inequality, crisis, modernisation. — Moscow: NISP, 2010. — 160 p.

4. Litvintseva G.P., Vavilova A.A., Voronkova O.V. Digital inequality of Russian regions: assessment and factors // Problems of Forecasting. — 2020. — No. 5. — P. 97–107.

5. Kislitsyna O.A. Measuring quality of life / well-being: international experience. — Moscow: IE RAS, 2016. — 62 p.

6. Naiden S.N., Belousova A.V. Social infrastructure and quality of life in Russian regions // Spatial Economics. — 2020. — Vol. 16, No. 1. — P. 107–131.

7. Bobkov V.N., Novikova I.V., Shichkin I.A. Inequality of quality and standard of living in territorial dimension // Economy of Region. — 2021. — Vol. 17, No. 1. — P. 33–47.

8. Grishina I.V., Polynev A.O. Comprehensive assessment of investment attractiveness of Russian regions. — Moscow: LENAND, 2018. — 160 p.

9. Surinov A.E. Statistics of population incomes. — Moscow: Finstatinform, 2001. — 239 p.

10. Rosstat. Regions of Russia. Socio-Economic Indicators — 2024. — Moscow: Rosstat, 2024.

11. Rosstat. Regions of Russia. Socio-Economic Indicators — 2023. — Moscow: Rosstat, 2023.

12. Rosstat. Regions of Russia. Socio-Economic Indicators — 2022. — Moscow: Rosstat, 2022.

13. Rosstat. Regions of Russia. Socio-Economic Indicators — 2021. — Moscow: Rosstat, 2021.

14. ROMIR. Russian Household Welfare Index. — Moscow: ROMIR, 2023. — URL: https://romir.ru.

Foreign-language sources

15. Sen A. Development as Freedom. — New York: Oxford University Press, 1999. — 366 p.

16. Stiglitz J.E., Sen A., Fitoussi J.-P. Mismeasuring Our Lives: Why GDP Doesn't Add Up. — New York: The New Press, 2010. — 176 p.

17. Diener E., Suh E.M., Lucas R.E., Smith H.L. Subjective well-being: Three decades of progress // Psychological Bulletin. — 1999. — Vol. 125, No. 2. — P. 276–302.

18. Nussbaum M.C. Creating Capabilities. — Cambridge: Harvard University Press, 2011. — 237 p.
19. Kaiser H.F. An index of factorial simplicity // Psychometrika. — 1974. — Vol. 39, No. 1. — P. 31–36.
20. Becker G.S. Human Capital. — 3rd ed. — Chicago: University of Chicago Press, 1994. — 412 p.
21. Alkire S., Foster J. Counting and multidimensional poverty measurement // Journal of Public Economics. — 2011. — Vol. 95, No. 7–8. — P. 476–487.
22. OECD. Handbook on Constructing Composite Indicators. — Paris: OECD Publishing, 2008. — 162 p.
23. OECD. Better Life Index. — Paris: OECD, 2023. — URL: https://www.oecdbetterlifeindex.org.
24. UNDP. Human Development Report 2023/2024. — New York: UNDP, 2024. — 340 p.
25. Nguefack-Tsague G., Klasen S., Zucchini W. On weighting the components of the Human Development Index // Journal of Human Development and Capabilities. — 2011. — Vol. 12, No. 2. — P. 183–202.
26. Booysen F. An overview and evaluation of composite indices of development // Social Indicators Research. — 2002. — Vol. 59, No. 2. — P. 115–151.
27. Decancq K., Lugo M.A. Weights in multidimensional indices of wellbeing // Econometric Reviews. — 2013. — Vol. 32, No. 1. — P. 7–34.